Homework 1

1. Consider the backwards heat equation modelled by the forward operator

$$Kx(t) = \int_0^\pi x(s)k(t,s)\mathrm{d}s \quad k(t,s) = \frac{2}{\pi}\sum_{n=1}^{\infty} e^{-Tn^2}\sin(nt)\sin(ns).$$

   (a) We know that K is compact in $L^2(0,\pi)$. Show that K has the singular system given by

$$x_j(t) = \sqrt{\frac{2}{\pi}}\sin(jt), \quad y_j(s) = \sqrt{\frac{2}{\pi}}\sin(js), \quad \mu_j = e^{-Tj^2}.$$

   (b) Consider the implementation of K by matrix $A \in \mathbb{R}^{M\times M}$ for $T = 0.05$. Compute the SVD of A. Compare in a plot the singular values of K and A, and in a separate plot, the singular vectors for A to the singular functions for K. Show by a numerical experiment that $Kx_j = \mu_j y_j$ for $j = 2, 4, 8$.

   (c) Fix the function

$$x(s) = s\sin(s)\exp((s-\pi/2)^2).$$

   Plot (using the discrete approximations) the functions x and $y = Kx$. Perturb y by adding Gaussian noise n of relative size $\|n\|_{L^2(0,\pi)}/\|y\|_{L^2(0,\pi)} = 2\%$ to obtain y^δ, and make a Picard plot of the singular values μ_j, the expansion coefficients $|(y^\delta, y_j)|$ and the reconstruction coefficients $|(y^\delta, y_j)|/\mu_j$.

2. Consider the Abel problem defined in $L^2(0,1)$ by

$$Kx(t) = \int_0^1 k(t,s)x(s)\mathrm{d}s, \qquad k(t,s) = \begin{cases} \frac{1}{\sqrt{(t-s)}}, & s < t \\ 0, & s > t. \end{cases}.$$

   With your a numerical discretization based on a quadrature rule, set up the matrix A corresponding to uniform $s-$ and $t-$meshes with $n = 1000$ points. Fix the function

$$x(s) = s^2 - \cos(2\pi s).$$

   Discretize the function x by a vector x, compute $y = Ax$ and add Gaussian noise of relative size $\delta = 10\%$ to obtain y^δ

   (a) Compute TSVD approximation x_k for well chosen values in $k \in [1, 100]$. Create a plot of reconstructions, and compare it to the ground truth (x) via a plot of the relative error $x - x_k/\|x\|_2$.

(b) Compute Tikhonov solutions x_α for well chosen regularization parameters $\alpha \in (10^{-4}, 10)$. Plot solutions and compare to TSVD.

(c) Try the above with a discontinuous function x.

3. Let K be a compact from Hilbert space X to Hilbert space Y with singular system $\{x_j, y_j, \mu_j\}$. Assume that K is injective. Let R_α be the regularization operator for K based on Tikhonov regularization written in the singular system for K as

$$x_\alpha = R_\alpha y = \sum_j \frac{\mu_j}{\mu_j^2 + \alpha}(y, y_j)x_j.$$

(a) Show that $\lim_{\alpha \to 0} x_\alpha = x$.
(Hint: use the expansion above to consider $\|R_\alpha K x - x\|^2$. Take the limit $\alpha \to 0$; be careful when interchanging the limit and summation.)

(b) Consider the deblurring model problem with solution x, matrix A, $y = Ax$, and $y^\delta = y + d$ with $\|d\| < \delta$. Choose the Tikhonov regularization parameter $\alpha(\delta)$ such that

$$\lim_{\delta \to 0} \alpha(\delta) = 0, \qquad \lim_{\delta \to 0} \frac{\delta^2}{\alpha(\delta)} = 0.$$

Demonstrate numerically that $\lim_{\delta \to 0} \|R_{\alpha(\delta)} y^\delta - x\| = 0$.
(Hint: Take for instance $\delta = 0.01 : 0.01 : 0.1$ and $\alpha(\delta) = \delta$.)

Important notes:

- Deadline for the HW1 is Friday October 11.
- All homeworks must be approved in order to qualify for the exam.
- Demonstrate in your solution your understanding of theory and computations.
- Each student should hand in a single pdf via Learn. The number of pages for the main report must not exceed 5. Code can be put in an appendix.
- Comment on all results: What do you see, and why is it reasonable (or unreasonable)?
- Make quality plots with proper names on axis, legends and captions. Zoom in on the relevant stuff.